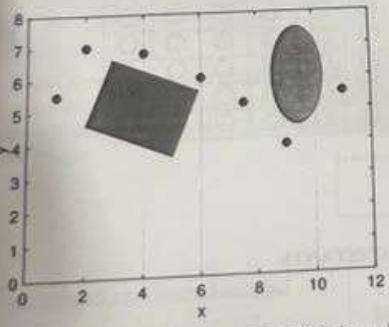
MULTIPLE-CHOICE QUESTIONS

1. How many fourth-degree polynomials pass through the points (1, 1), (2, 8), (3, 27) and (4, 64)?

   a. 1
   b. 2
   c. 0
   d. Infinitely many
   e. None

2. A robot has to follow a path that passes through seven planning points in order to avoid two obstacles, as shown in the plot. To find the shortest path that is also smooth, which of the following solution strategies would you recommend?



   a. Pass a sixth degree polynomial through the data
   b. Pass linear splines through the data
   c. Pass cubic splines through the data
   d. Use least squares to fit a second degree polynomial
   f. Pass a quadratic interpolation through all data

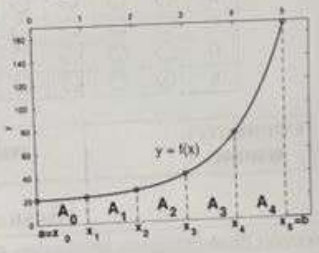
3. If data is known to be very precise, which method should be used to fit a curve?

   a. NR
   b. Bisection
   c. Regression
   d. Secant
   e. Spline Interpolation

4. In cubic spline interpolation,

   a. only the first derivatives of the splines are continuous at the interior data points.
   b. only the second derivatives of the splines are continuous at the interior data points.
   c. the first and the second derivatives of the splines are continuous at the interior data points.
   d. only the third derivatives of the splines are continuous at the interior data points
   e. there is no requirement for the third derivatives of the splines.

5. The integral $\int_0^5 f(x)\,dx$ is approximated using the left rectangle rule (A_L), right rectangle rule (A_R) and midpoint rule (A_M) for the function f(x) shown in the plot. Which of the following inequalities must be TRUE?

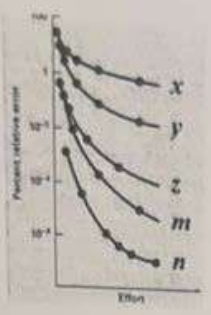


   a. $A_L < A_M < A_R$
   b. $A_M < A_L < A_R$
   c. $A_R < A_M < A_L$
   d. $A_M < A_R < A_L$
   e. $A_M < A_L = A_R$

6. Which one is **false**?

   a. Error in Taylor series expansion with only first derivative used, is $O(h^2)$
   b. Forward finite-divided difference approximation (FDDA) does not yield better results than backward one.
   c. Forward and backward FDDA methods have similar accuracy.
   d. Forward FDDA: $\frac{df(x)}{dx} \approx \frac{f(x+\Delta x)-f(x+\Delta x)}{2\Delta x}$
   e. Backward FDDA: $\frac{df(x)}{dx} \approx \frac{f(x)-f(x-\Delta x)}{\Delta x}$

7. Comparison of percent relative error versus computational effort for 1st through 5th RK methods is given in the figure. Which one corresponds to the method shown by **x**?



a. RK-4th
b. Euler
c. Heun
d. Butcher
e. Taylor

8. For $u = f(x)$ and $v = g(x)$, which of the followings is *true*?

a. If $y = uv$, then $\frac{dy}{dx} = u\frac{du}{dx} + v\frac{dv}{dx}$
b. If $y = uv$, then $\frac{dy}{dx} = u\frac{dv}{dx} - v\frac{du}{dx}$
c. If $y = u/v$, then $\frac{dy}{dx} = \frac{v\frac{du}{dx} - u\frac{dv}{dx}}{v}$
d. If $y = v/u$, then $\frac{dy}{dx} = \frac{u\frac{dv}{dx} - v\frac{du}{dx}}{u^2}$
e. If $y = u^v$, then $\frac{dy}{dx} = u^v\left[\frac{dv}{dx}\ln(u) - \frac{v\frac{du}{dx}}{u}\right]$

9. Which one is *true* for numerical methods of ordinary differential equations (ODE)?

a. Heun method give exact result when the solution of ODE is cubic.
b. For 2nd order RK method, there are 3 equations and 3 unknowns.
c. Local and global errors of Euler method are 1st order.
d. Midpoint method is 1st order accurate.
e. Ralston method is a 3rd order RK method

10. Given $y = \cos\left(e^{\cos(x)}\right)$, what is $\frac{dy}{dx}$?

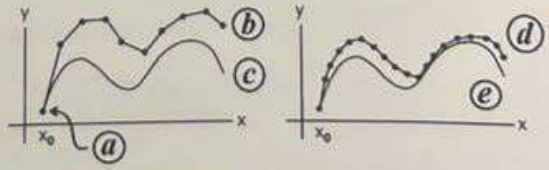
a. $-\sin(x)\sin\left(e^{\cos(x)}\right)$
b. $\sin\left(e^{\sin(x)}\right)$
c. $-\sin(x)\,e^{\cos(x)}\cos\left(e^{\cos(x)}\right)$
d. $\sin(x)\,e^{\cos(x)}\sin\left(e^{\cos(x)}\right)$
e. $-\sin(x)\,e^{\cos(x)}\sin\left(e^{\cos(x)}\right)$

11. Using the backward divided difference approximation with a step size of 0.2, the derivative of the function (given in below table) at x = 2.0 is

x	1.8	2.0	2.2	2.4	2.6
$f(x)$	6.0496	7.3890	9.0250	11.023	13.464

a. 9.206
b. 8.180
c. 7.438
d. 9.990
e. 6.697

12. Below plots give the unknown y(x) and numerical aproximations with a step size and an initial condition by Euler' Method. Which one is **false**?



a. b corresponds to an aproximation.
b. d corresponds to approximation with bigger step size.
c. d is a better apprimation than b.
d. a corresponds to the initial condition.
e. c corresponds to the unknown y(x).

CLASSIC QUESTIONS

1. **(20 p)** Using the Euler Method calculate the approximate solution of the IVP given as $y' + 1.25y = 1 + 0.5e^t$ with $y(0) = 1$ having steps h=0.5 at the points $t = (0.5, 1.0, 1.5, 2.0, 2.5)$. Compare the values you found with the exact solution given as $y_{exact} = (10e^t - 6e^{-1.25t} + 36)/45$ and calculate the percent error. Use 5 decimal places in calculations.

2. **(20 p)** Fit $f(x)$ to $F(x) = ae^{bx}$ using the table below and the least squares method. Use 4 decimal places in calculations.

x	1.50	2.30	3.70	4.20	5.00
$f(x)$	1.62	2.40	4.85	7.64	10.10

Note: Show what you have done in your calculations at every step in detail. Otherwise, even if your calculation result is correct, it will not be graded.